

NGUYEN THANH PHUONG

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SUMMARY

Final-year Computer Science student focused on AI Engineering, computer vision, and machine learning. Experienced in PyTorch, BasicSR, image super-resolution, model evaluation, data analysis, and technical documentation.

WORK EXPERIENCE

Vietnam AI Software Solution - Business Analyst Intern

Nov 2025 - Feb 2026

- Gathered and analyzed business requirements from school stakeholders, including administrators, teachers, and staff.
- Translated business needs into functional specifications, user stories, and system workflows for the development team.
- Worked closely with developers to clarify requirements and prepared project documentation, reports, and presentation materials.

Saigon High-Tech Park - IT Helpdesk Intern

Sep 2025 - Nov 2025

- Provided technical support for hardware and software issues for internal users.
- Assisted in system setup, configuration, and troubleshooting to support stable daily operations.

PROJECTS

AdaptiveCATANet for Lightweight Image Super-Resolution

Tools: PyTorch, BasicSR, CUDA, Google Colab, Computer Vision

- Proposed an improved CATANet-based architecture for lightweight Single Image Super-Resolution using Adaptive Token Allocation.
- Designed AdaptiveTAB with a Token Predictor to combine K=16 and K=64 token branches through Soft-Selection.
- Integrated Channel Attention, SiLU, Charbonnier Loss, and Sparsity Loss to improve representation and training stability.

Fair Comparison of MLP and KAN

Tools: PyTorch, KAN, NumPy, Matplotlib

- Built a fair experimental framework to compare MLP and KAN on function approximation and MNIST classification.
- Controlled model settings, optimizer, learning rate, batch size, and training setup to reduce evaluation bias.
- Analyzed learning curves, FLOPs, parameter count, convergence behavior, and model trade-offs.

SKILLS

- **AI / Machine Learning:** Regression, Classification, CNN, Transformers, Model Evaluation
- **Computer Vision:** Image Super-Resolution, Attention Mechanisms, PSNR, SSIM
- **Programming & Libraries:** Python, SQL, C++, PyTorch, TensorFlow, scikit-learn, pandas, NumPy
- **Tools:** Git/GitHub, Jupyter Notebook, Google Colab, BasicSR, Streamlit
- **Databases:** MySQL, MongoDB
- **Languages:** Vietnamese (Native), English (TOEIC 750)

EDUCATION

Open University

Bachelor of Computer Science

Expected in 8/2026

GPA: 3.2/4.0

HONORS & AWARDS

Consolation Prize - University-level Scientific Research Competition

Project: AdaptiveCATANet for Lightweight Image Super-Resolution 2025

LANGUAGES

English: TOEIC 700

GITHUB LINKS

<https://github.com/char704>